

Context

The Data Science Project will verify the learning outcome of several courses of Master 1st year, specifically Data Preparation, Data Warehouse Modeling, Data Mining and Statistics, but also Professional Communication and background knowledge on project management.

Upon successful completion of this course, the students must be able to:

- Model, design and implement a data warehousing system, based on a need-driven workload.
- Integrate data from several types of data sources, including (but not limited to) relational databases, flat files, web data, semi-structured data, documents and maps.
- Prepare raw input data, and process it appropriately to provide suitable input for a wide range of analysis tasks.
- Understand the theoretical background and parameters of the main data mining algorithms.
- Critically evaluate and select appropriate algorithms and tools.
- Perform query and analysis tasks, interpret the results and report outputs appropriately.

Several specific topics are proposed to students. Each topic describes a set of user needs and an available set of data sources. An example of topic is provided at the end of this document. Students may choose among the proposed topics or provide a new one (to be discussed and approved by advisors).

Learning outcome and objectives

This project aims at the design and implementation of a business intelligence system supporting data analysis.

The expected outcome is a tool that should allow to: (1) integrate source data in a data warehouse, (2) prepare the data for analysis, and (3) perform in-depth analyses over these data.

The project detailed objectives are:

1. Data integration: automation of data integration by means of ETL scripts, identification of problems to be corrected, diagnostic of data quality before and after the integration.
2. Data warehouse design and implementation: modelling, design and implementation of a data warehouse in the presence of a need-driven workload. Query response time must allow for interactive navigation at various levels of granularity.
3. Interactive analysis of warehouse data as a basis for future analyses, including navigations, different types of visualizations, and relevant conclusions.
4. Data mining analysis of warehouse data, including selection of appropriate algorithms, and interpretation and reporting of results.

Organization and milestones

Students should organize in teams of 3 or 4 members. Teams cannot merge regular students and apprentices.

Each team must propose its original solution to the project subject. Each team must clearly enounce task distribution and designate a leader for each project phase (each student must ensure the leadership of at least one phase).

It is not mandatory to use a big number of datasets, nor solving a large amount of user needs, but the larger the data warehouse, the more requirements are covered, the more data mining methods are deployed, the more analysis are performed, and the better it is. Nevertheless, the project must satisfy a list of constraints that ensures its challenging character.

Each project phase concludes with a report, a video and/or a demo. In addition, each semester concludes with an overall report and an oral presentation. Reports, videos and slides must be uploaded on the Celene page of the course (should the video be too big to be uploaded, a text file with the link to the video is to be uploaded instead).

Organization is different for regular students (hereafter FI) and apprentices (hereafter FA): All teams are asked to describe and model the whole set of requirements of their topic. However, while FI teams will implement the whole data warehouse, FA teams will choose a subset of requirements to implement and report.

The following table shows project phases for the first semester; the ones for the second semester will be presented in January:

Dates	Phase	Evaluated work
1/9 – 11/9	Study: choice of topic, workload, data extraction	Report (11/9) and Q&A session (15/9)
15/9 – 3/10	Modeling: conceptual schemas, data dictionary, formal workload	Report (3/10) and Q&A session (6/10)
15/9 – 6/11	Loading: architecture, logical schemas, data profiling, ETL	Report (6/11) and demo (10/11)
10/11 – 11/12	Reporting: interactive analysis	Report (27/11 FA - 11/12 FI) and oral presentation with demo (1/12 FA – 15/12 FI)

Intermediate reports should contain technical documentation in order to complement live demos. However, final reports should be descriptive, with examples and explanations in addition to technical diagrams.

More precisely, the expected outcomes of each phase of first semester are the following:

- Study. Report: description of the selected topic, detailed study of user needs resulting in a preliminary workload, description and size of data sources (with links). Project management annex (work distribution, Gantt diagram, advancement indicators).
- Modeling. Report: DW conceptual schemas (in DFM), data dictionary, formalization of preliminary workload phrased in Golfarelli and Rizzi's abstract language, prototype of reports for illustrating the workload queries. Project management annex.

- Loading. Report: data warehouse architecture, logical schemas, data profiling, preliminary study of data distribution, illustration of the main quality issues and corrective actions (transformation, cleaning, discretization, etc.), quantitative analysis of quality problems before and after correction, high-level (code-free) ETL workflows. Project management annex. Demo: overview of source schemas, illustration of main transformations, overview of the loaded data warehouse with testing queries.
- Reporting. Report: Summary of project results, description of some analyses for solving user needs. Link for accessing the DW. Project management annex. Demo: DW interactive querying illustrating the answer to users' needs.

Constraints

Each team is free to define the specific topic of their project, provided that the following conditions are satisfied:

- Workload: The query workload should contain at least 2 users' needs per student. Needs must be defined in a fuzzy manner, pertaining to several dimensions and measures, such that answering the needs requires interactive exploration.
- Machine learning needs: The mining workload should contain 1 or 2 users' needs per student (i.e. 1 for FA students, 2 for FI students). It is recommended to have at least 1 supervised and 1 unsupervised task.
- Data sources: At least 2 different sources must be integrated (3 sources may be considered for FI teams). They must be chosen to illustrate integration issues. Data sources should be open access.
- Quality issues: Sources must be such that data quality issues are illustrated, needing at least 2 different types of corrective actions (at least 4 types is recommended for FI teams).
- Data volume: the data volume of the data warehouse must allow interesting analysis of the underlying data. 500 facts is the minimum necessary for supervised tasks.

Example of project topic

A previous topic was related to a challenge initially proposed during the French speaking conference EGC'2017 (see <http://egc2017.imag.fr/defi/> for more details). The challenge has been initiated by the LIG Computer Sciences Lab of Grenoble, the city hall of Grenoble and Big Datest, a company that is specialized in predictive analysis. All together they built an open data set related to the diseases of the trees around the city of Grenoble.

Source datasets

The EGC'2017 provides two data sets (around 10K tuples) for learning about trees. Both data sets contain:

- A file describing trees, planting, diagnostics and treatments.
- A file providing geo-coordinates for each tree.
- A file indicating tree health.

A third data set, for testing, contains description and geo-coordinates files, but no health indication.

Each team had to use at least one additional data source, of their choice, for complementing EGC'2017 data sets. Open data sources are encouraged.

User needs

Based on EGC'2017 challenge description, several user needs were identified:

- ❖ Have better understanding about Grenoble's green resources (tree stock), understand their evolution and provide guidelines to ease their maintenance.

Some examples of studies:

1. General statistics about trees (ex. number of trees, average age, average diameter) according to several analysis axes (ex. variety, location...).
2. Evolution of such statistics
3. Georeference of trees, allowing the search of specific genus and species.
4. Study of a particular variety or species (presence by location, evolution, development).
5. Aging of trees, comparing planting of new areas and renewal, by location and genus. Identification of neglected areas.
6. Influence of external features (ex. weather, pollution, relief, presence of industries, economical situation, investments...) in tree stock.

- ❖ Have better understanding about tree diseases.

Some examples of studies:

1. General statistics about sick trees and treatments according to several analysis axes.
2. Georeference of locations concentrating sick trees.
3. Relation among various features and tree disease. Note that such features may include attributes in EGC'2017 data sets (ex. location, time), calculated attributes (ex. age) but may also include external features (ex. whether, pollution, relief, presence of industries...).

- ❖ Predict tree disease

Some examples of studies:

1. Predicting if a tree has a disease or not.
2. Predicting which part of the tree suffers from a disease. Considered tree parts are: collet ("collet"), crown ("houppier"), root ("racine") and trunk ("tronc"). Note that several parts can be sick or none.
3. Conducting any data mining task you may think of, trying to extract and learn as much.

Each team had to customize and enrich these needs by learning about the topic and application domain, integrating additional data sources, and choosing related topics of their preference.